SYSTEM ARCHITECTURE & PROJECT DESIGN SPECIFICATION

AI Weather Wise

1. ARCHITECTURE

```
Client Requests → API Gateway / Controller → Weather Service & Model → MongoDB /
Gemini AI → Adaptive Forecast JSON
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The MVC pattern cleanly separates client routing, weather telemetry controllers, domain business logic, data models, and the dedicated Gemini AI inference service for intelligent weather forecasting and personal advisory generation.

2. LAYER DESIGN

Client & Frontend Layer

- Web Weather Dashboard & Interactive Maps
- Mobile Weather App (iOS / Android)
- Alerts & Push Notification Widget

Middleware & Security

- JWT Authentication & API Key Management
- Rate Limiting (OpenWeather & Local Cache)
- Geofencing & CORS Security Policies

Routes & Controllers

- **Routes:** Auth, Forecast, Location History, AI Advice
- **Controllers:** Weather telemetry, Alert triggers, Advisory engine, Cache manager

Model & Service Layer

- Mongoose models (User, Location, WeatherCache)
- OpenWeather / NOAA Data Ingestion Pipelines
- Gemini AI Contextual Recommendation Engine

3. DATABASE DESIGN

MongoDB document collections include **Users**, **SavedLocations**, and **WeatherSnapshots**.
`SavedLocations.userId` references `User._id` for customized user updates.
`WeatherSnapshots.locationId` indexes geo-coordinates to enable fast caching and minimize external API calls.
Supports one-to-many relationship mapping across historical weather activity.

4. API DESIGN

Module	HTTP Method & Endpoint	Description
Auth	`POST` /api/auth/register `GET` /api/auth/profile	User registration & authenticated preferences management.
Weather Data	`GET` /api/weather/current `GET` /api/weather/forecast	Fetch real-time weather, radar imagery, and multi-day forecasts.
Locations	`POST` /api/locations `PUT` /api/locations/:id	Manage saved user locations, favorite alerts, and default coordinates.
AI Engine	`POST` /api/ai/weather-advisory	RAG-driven personalized weather summaries, outfit & activity recommendations.
AI Engine	`POST` /api/ai/severe-alert-summary	Automated triage of extreme weather alerts with plain-language safety guidance.